



## What is Packet Sniffing ?

Last Updated : 23 Mar, 2023

When any data has to be transmitted over the computer network, it is broken down into smaller units at the sender's node called *data packets* and reassembled at receiver's node in original format. It is the *smallest unit* of communication over a computer network. It is also called a block, a segment, a datagram or a cell. The act of capturing data packet across the computer network is called **packet sniffing**. It is similar to as wire tapping to a telephone network. It is mostly used by *crackers and hackers* to collect information illegally about network. It is also used by *ISPs, advertisers and governments*. **ISPs** use packet sniffing to track all your activities such as:

- who is receiver of your email
- what is content of that email
- what you download
- sites you visit
- what you looked on that website
- downloads from a site
- streaming events like video, audio, etc.

**Advertising agencies** or internet advertising agencies are paid according to:

- number of ads shown by them.
- number of clicks on their ads also called PPC (pay per click).

To achieve this target, these agencies use packet sniffing to *inject advertisements* into the flowing packets. Most of the time these ads *contain malware*.

**Government agencies** use packet sniffing to:

- ensure security of data over the network.
- track an organisation's unencrypted data.

**Packet Sniffer** – Packet sniffing is done by using tools called *packet sniffer*. It can be either *filtered or unfiltered*. Filtered is used when only *specific data packets* have to be captured and Unfiltered is used when *all the packets* have to be captured. WireShark, SmartSniff are examples of packet-sniffing tools.

**How to prevent packet sniffing –**

- Encrypting data you send or receive.
- using trusted Wi-Fi networks.
- Scanning your network for dangers or issues.

**Advantages:**

- **Network troubleshooting:** Packet sniffing can be used to identify network problems by examining the packets and identifying issues such as network congestion, packet loss, or improper configuration.
- **Security analysis:** Packet sniffing can be used to detect and analyze security threats, such as network intrusions, malware infections, or unauthorized access attempts.
- **Network optimization:** Packet sniffing can be used to optimize network performance by identifying bottlenecks and optimizing the network configuration.

- **Protocol analysis:** Packet sniffing can be used to analyze network protocols and identify areas where they can be improved or optimized.

#### Disadvantages:

- **Privacy violations:** Packet sniffing can be used to intercept sensitive information, such as passwords, credit card numbers, or personal information, which can be used for malicious purposes.
- **Legal issues:** In many jurisdictions, packet sniffing is illegal without the express consent of all parties involved in the communication.
- **Resource usage:** Packet sniffing can consume a significant amount of system resources, especially if large amounts of network traffic are being analyzed.
- **Complexity:** Packet sniffing can be a complex process, requiring specialized knowledge and tools to analyze network traffic effectively.

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